



# VITAMIN B COMPLEX

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# VITAMIN B COMPLEX

- Vitamin B complex is a group of water soluble vitamins.
- That play essential role in energy metabolism, enzyme function, red blood cell production.
- It include eight B vitamins B1(thiamine), B2 (riboflavin), B3 (niacin), B5 ( pantothenic acid), B6 (pyridoxine), B7 (biotin), B9 (folic acid), and B12 (cobalamin).
- B vitamins are not stored in the body except vitamin B12, they must be consumed regularly through diet or supplements.



# VITAMIN B1 (THIAMINE)

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- Biological action: Thiamine pyrophosphate is the active form of thiamine.
- Acts as a cofactor for transketolase and pyruvate dehydrogenase in glycolysis.
- It is also imp for energy generation in the citric acid cycle that is highly active in the heart and brain.



## REQUIREMENTS:

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- <6 months: 0.2mg
- 6-12 months: 0.3mg
- 1-3 years: 0.5mg
- 4-8 years: 0.6mg
- 9-13 years: 0.9mg
- >14 years: 1.2mg



- **DIETARY SOURCES:**

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- Found in human milk, cow milk, unpolished grain, egg, organ meats, (liver, kidney), and legumes
- Sensitive to heat, pasteurization and sterilization.



## DEFECIENCY:

- Thiamine deficiency result in beriberi mainly affect people who consume diets based on polished rice.
- 3 forms of beriberi:
- **DRY BERIBERI:** Peripheral neuritis with irritability, paralysis of lower limb, loss of deep tendon jerks, muscle wasting, No edema is seen in dry beriberi.
- **WET BERIBERI:** Congestive cardiac failure and edema of legs, face, and trunk are main features.

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- **INFANTILE BERIBERI:** Affects breastfed infants of thiamine deficient mothers or very low thiamine intakes. Characteristics by cardiomegaly, cyanosis, dyspnea.
  - **WERNICKE KORSAKOFF SYNDROME:**
  - Also called cerebral beri beri
  - Severe deficiency of thiamine causes this syndrome
  - Seen in chronic alcoholics, characterised by mental confusion, ataxia, and ocular abnormalities.
  - Can also occur in thiamine deficient children.



## • DIAGNOSIS:

- Thiamine deficiency may be suspected in all cases of malnutrition. The diagnosis is confirmed by measuring 24 hours urinary thiamine excretion or red blood cell transketolase activity .

## • TREATMENT:

- Treatment with thiamine leads to resolution of neurologic and cardiac symptoms within 24-48 hours.
- Mild beriberi- 5mg/day of thiamine
- Severe case- 10 mg IV twice daily..



# RIBOFLAVIN (VITAMIN B2)

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- Riboflavin is a constituent of two coenzyme involved in redox reactions of cellular respiration.
- Flavin adenine dinucleotide ( FAD ) and flavin mononucleotide ( FMN )
- Riboflavin coenzymes also participate in tissue protein building.



## REQUIREMENTS:

- <1years : 0.3mg
- 1-3 years: 0.5mg
- 4-8 years: 0.6 mg
- 9-14 years: 0.9 mg
- >14 years :1.2 mg

### Dietary source

- Meat, milk, eggs, and cereals are good source of riboflavin.
- It is resistant to oxidation and heat and is not destroyed by pasteurisation.

## DEFICIENCY:

- It causes ariboflavinosis.
- It takes 1-2 months to develop.
- Features include photophobia, glossitis, cheilosis, angular stomatitis, seborrheic dermatitis and corneal vascularization.
- **TREATMENT:**
- Children-3-10mg oral riboflavin daily for several weeks.
- Infants-1mg oral riboflavin daily..





# NIACIN (VITAMIN B6)

- Nicotinic acid and Nicotinamide are both referred to as niacin.
- Niacin is a part of NAD and NADP involved in redox reaction and cellular respiration.
- SOURCES:
- Meat, eggs, legumes, peanuts , milk etc.



## Daily requirements:

- <1 year: 2-4mg
- 1-3 years: 6 mg
- 4-8 years: 8mg
- 9-13 years: 12mg
- >13 years: 16 mg

## DEFICIENCY:

- It leads to pellagra, which is characterised by 3 d's : Dermatitis, Diarrhea, Dementia.
- And if not treated it may lead to 4th d- death.
- The cutaneous lesions consist of symmetrical pigmented rashes in body parts exposed to sunlight, especially in the neck (casal's necklase).





- Treatment:

- 50-300 mg/day( 10 times more than dietary intake).
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# PANTOTHENIC ACID( VITAMIN B5)

- Pantothenic acid is a part of coenzyme A , required for the metabolism of fatty acid, proteins and carbohydrates in citric acid cycle and fatty acid synthesis.
- Daily requirement:
- Infants: 2-3 mg
- Children: 3-5mg
- Deficiency: rare and includes burning feet,insomnia and Gastrointestinal syndrome.



# PYRIDOXINE(VITAMIN B6)

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- Pyridoxin is converted into coenzyme pyridoxal phosphate, which is involved in amino acid metabolism, synthesis of sphingolipids, neurotransmitter and hemoglobin production.
- **SOURCE:**
- Meat, whole grain, cereals, vegetables, soya beans, egg yolk.



- **DEFICIENCY:**

- Peripheral neuropathy, refractory seizures, dermatitis and microcytic anemia.

- **TREATMENT:**

- Pyridoxine 10-15 mg /day is given to patients on INH(isoniazid) to prevent peripheral neuropathy.
- Pyridoxine 100 mg IM is injected to prevent refractory seizures.



# BIOTIN (VITAMIN B7)

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- Biotin is a coenzyme for many carboxylase enzymes involved in gluconeogenesis, amino acid synthesis, and fatty acid synthesis.
- SOURCES:
  - Liver, egg yolk, milk , yeast extracts and meat.
- DAILY REQUIREMENTS:
  - 20-30 micrograms/day.



- **DEFICIENCY:**

- Anorexia, vomiting, dry scaly dermatitis, glossitis , developmental delay, and seizures.
- High consumption of raw egg also causes biotin deficiency.
- It is also seen in genetic defects affecting the activity of carboxylase synthetase, biotinidase etc.
- **TREATMENT:**
- 5-10 mg/day orally for 2-3 weeks .



REFERENCE: O.P GHAI , SATHYANARAYANA

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**THANK YOU**